

Proposition 2.2. *Let $V \neq \{0\}$ be an ordered normed space whose cone has an interior point v . Then v is an order unit and the order unit norm $\|\cdot\|_v$ is weaker than the original norm $\|\cdot\|$ on V . Moreover, every positive functional on V is continuous with respect to both norms.*

Note that the two norms in Proposition 2.2 need not be equivalent, see, e.g., [6, Example 1.5.12]. For a detailed discussion in ordered Banach spaces, see [4, Proposition 2.11].

For an ordered vector space (V, V_+) , a set S' of positive linear functionals from V to $\mathbb{R}$ is said to *determine positivity* if, for every $v \in V$, one has $v \geq 0$ if $\varphi(v) \geq 0$ for all $\varphi \in S'$. Note that, if V is an order unit space and $\Sigma \subseteq V'_+$ is given as in (2.1), then $\text{ext } \Sigma$ determines positivity.

We will make use of the following two simple propositions.

Proposition 2.3. *Let X be an ordered normed vector space and let $S' \subseteq X'_+$. The following are equivalent:*

- (i) *The set S' determines positivity, i.e., for every $x \in X$, the inequality $\langle x', x \rangle \geq 0$ for all $x' \in S'$ implies that $x \geq 0$.*
- (ii) *The positive-linear hull of S' is weak-* dense in the dual wedge X'_+ .*

Proof. “(i) $\Rightarrow$ (ii)” We show the contraposition. Let $C' \subseteq X'_+$ denote the weak-* closure of the positive-linear hull of S' and assume that there exists a point $x'_0 \in X'_+$ that is not in C' . By the Hahn–Banach separation theorem in the locally convex space X' with the weak-* topology, there exists a weak-* continuous functional $x'' : X' \rightarrow \mathbb{R}$ and a number $\gamma \in \mathbb{R}$ such that $\langle x'', x' \rangle \geq \gamma > \langle x'', x'_0 \rangle$ for all $x' \in C'$. Due to the weak-* continuity of x'' , there exists an element $x \in X$ such that x'' is the point evaluation at x , see [13, § IV.1.2 on pp. 124–125]. It follows that $\langle x, x' \rangle \geq \gamma > \langle x, x'_0 \rangle$ for all $x' \in C'$. Since $0 \in C'$, we obtain $\gamma \leq 0$. Therefore, x is not in X_+ . As C' is a wedge, for every $x' \in C'$ we have $\langle x, x' \rangle \geq 0$, hence C' does not determine positivity.

“(ii) $\Rightarrow$ (i)” If $x \in X$ satisfies $\langle x', x \rangle \geq 0$ for all $x' \in S'$, then it follows from (ii) that the same inequality holds for all $x' \in X'_+$. As X_+ is closed, this implies that $x \in X_+$. $\square$

The following is a primal version of Proposition 2.3. As its proof is similar, we omit it.

Proposition 2.4. *Let X be an ordered normed vector space and let $S \subseteq X_+$. The following are equivalent:*

- (i) *The set S determines positivity of functionals in the sense that, for each $x' \in X'$, the inequality $\langle x', x \rangle \geq 0$ for all $x \in S$ implies that $x' \geq 0$.*
- (ii) *The positive-linear hull of S is weakly dense (equivalently, norm dense) in X_+ .*

If the cone V_+ in an ordered vector space X has a base D , then $x \in D$ is extremal if and only if x is an extreme point of D , see [6, Lemma 1.5.19]. The following result by Krein and Milman is given, e.g., in [13, §II.10.5 on p. 68].

Proposition 2.5. *Let V be a locally convex Hausdorff space. Let $\Sigma \subseteq V$ be compact and convex and let $S' \subseteq \Sigma$. If $\Sigma = \overline{\text{conv}(S')}$ then $\text{ext } \Sigma \subseteq \overline{S'}$.*